SemLake: Zero-Copy Branching and Three-Way Semantic Merging for Data Lakehouses

Predicate-Level Conflict Verification over Deletion Vectors and Extensions to Agentic Workloads

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ABSTRACT

Cloud data lakehouses support zero-copy branching by sharing immutable Parquet manifests, yet their merge mechanisms remain tied to physical storage units. Current systems check conflicts at file or row-identifier granularity: edits to disjoint rows in a shared file trigger false aborts or full-file copy-on-write rewrites, whereas cross-file read-write hazards, such as a data delete concurrent with an aggregate query, merge silently and corrupt results. Meanwhile, programming-language semantic merges operate on in-memory syntax trees and database predicate locks target short interactive transactions; neither fits billion-row tables on object storage, where reading data files at merge time costs prohibitive I/O and API calls.

We present SemLake, a lakehouse storage and catalog architecture for predicate-level three-way semantic merging without rewriting base Parquet files. SemLake decouples metadata-time conflict verification from physical reconciliation: each commit records a compact summary with SQL read/write predicates, schema actions, deleted rows' pre-images, and modified rows' fork/tip values, while physical updates compile into bitwise unions of Roaring deletion vectors. Under single-machine academic constraints, we combine physical Parquet/Roaring I/O, DuckDB scans at SF 1 and SF 10, a live 1,000-file Amazon S3 run, native Dolt/`delta-rs` checks, and multi-agent simulation, showing that: (1) two-tier verification over commit summaries catches all declared read-delete, read-update, foreign-key, and domain conflicts across 34 scenarios and 77 TPC-H branch pairs (1.00 recall and precision); (2) Roaring DV unions avoid false shared-file aborts with +0.9% localized SF 1 scan overhead; (3) merges issue 7 S3 requests at 10K files (1.76 s vs. 472 s for copy-on-write on live S3 at 1K files); and (4) group commit scales a 64-agent swarm to 12.1 commits/s with zero committed hazards.

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The source code, data, and/or other artifacts have been made available at https://github.com/WanderingSnail/semlake_milestone2.

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1 INTRODUCTION

Cloud data lakehouses [6, 21, 23, 40] decouple compute engines from cloud object storage (such as AWS S3) and expose transactional table snapshots over immutable Parquet files. To isolate data engineering jobs, write-audit-publish (WAP) quality gates, regulatory data scrubs, and collaborative analytics, dataset versioning systems [7, 8, 22, 29, 39] and zero-copy lakehouse catalogs [18, 19, 30, 36] allow users to fork a table snapshot in milliseconds by copying a metadata pointer, execute transformations on an isolated branch, and merge the verified branch back into `main`.

Creating a branch on cloud object storage is cheap because immutable Parquet files are shared across snapshots [8]. Merging two diverged table branches, however, lacks a sound correctness model at lakehouse scale. In production data lakehouses, concurrent branches interact across four recurring workload patterns:

- W_1 **(Shared-file disjoint updates):** Branches update or delete non-overlapping rows in the same multi-hundred-megabyte Parquet files.
- W_2 **(Hotspot key collisions):** Branches concurrently modify overlapping keys under varying contention density ρ, requiring structured isolation without abort storms.
- W_3 **(Semantic dependencies):** One branch deletes or modifies rows (e.g., a regulatory purge or data scrub) that a concurrent branch read to compute an aggregate or assert an invariant, writing its output to a disjoint file.
- W_4 **(Schema evolution and manifest scale):** One branch evolves the table schema while a concurrent branch appends data across deep manifests and multi-hop commit DAGs.

State-of-the-art open table formats and versioned databases evaluate conflicts solely at physical file granularity or row-identifier granularity, failing across W_1–W_4. Under file-level optimistic concurrency control (B_1) [6], W_1 triggers false conflict aborts, while copy-on-write row merge (B_2) [36] and tuple-dictionary versioning (B_3) [22, 29] incur severe write amplification or $O(N_{\text{files}})$ S3 requests on W_4. On W_2, existing engines either abort the entire branch (B_1–B_3) or silently overwrite colliding cells via last-writer-wins CRDTs (B_4) [17]. Worse, as shown in Figure 1, all file-, row-, and cell-level engines (B_1–B_4, Dolt [14]) fail on W_3: when a compliance branch T_A purges VIP records in a base file while an analytics branch T_B aggregates revenue over those same records into a new summary file, the modified file sets and row IDs are disjoint, causing existing systems to commit a corrupted snapshot.

In the programming-languages (PL) community, three-way structured and semantic merges have been formalized using verification conditions [34], version-space algebras [43], AST matching [42],

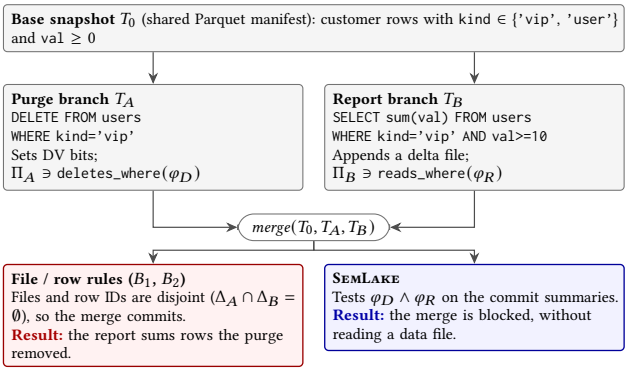


Figure 1: The lakehouse branch-merge dilemma on W_3. A compliance purge branch (T_A) and an analytical report branch (T_B) touch disjoint files and disjoint row identifiers. File- and row-level engines commit both branches silently, whereas SemLake intercepts the predicate conflict ($\varphi_D \wedge \varphi_R$) purely from commit metadata.

and post-merge static analysis or unit testing [12, 13]. However, PL techniques operate on abstract syntax trees measured in kilobytes. Lakehouse tables span billions of rows across thousands of cloud objects, where scanning data files at merge time incurs prohibitive S3 API latencies and byte transfer charges [15]. Testing whether a write touches a row selected by a reader's predicate goes back to predicate locks [16], precision locks [24], and optimistic read/scan validation [25, 26] for short transactions sharing one database. What is missing is a way to apply this check after the fact, between long-lived branches, without reading data files. We therefore evaluate lakehouse branch merging along four system metrics across W_1–W_4:

- M_1 **(Semantic correctness on W_1–W_4):** 100% recall on cross-branch semantic (W_3) and schema (W_4) conflicts, hotspot collision isolation (W_2), and 0% false aborts on shared files (W_1).
- M_2 **(Low-latency merge metadata work on W_2, W_4):** Scale three-way diff and merge CPU time with branch delta size rather than total file count (W_4), DAG depth (W_4), or contention ρ (W_2).
- M_3 **(Cloud cost efficiency on W_1, W_4):** Bound S3 API calls to $O(\Delta_{\text{manifest}})$ and maintain WAF ≈ 1 without full-file CoW rewrites.
- M_4 **(Bounded post-merge read overhead on W_1):** Preserve high-throughput analytical scans over merged snapshots prior to compaction.

To meet M_1–M_4 together, SemLake combines three mechanisms. First, for *zero-copy physical reconciliation* ($W_1, W_4 \rightarrow M_3, M_4$), SemLake represents row updates via positional Roaring deletion vectors (DVs) [5, 31] and delta manifests, compiling shared-file updates into bitwise bitmap unions and a single delta Parquet append. Second, for *metadata-time semantic and schema verification* ($W_3, W_4 \rightarrow M_1, M_2$), each commit writes a compact JSON summary capturing row operations, schema changes, deleted rows' pre-images, modified rows' fork/tip values, and SQL predicates from `SELECT` and `DELETE` statements. At merge time, SemLake finds the lowest common ancestor (LCA) via generation-pruned search, replays non-merge commits since the fork, checks static predicate satisfiability in metadata time, and, when static analysis cannot rule out an overlap, evaluates the predicate against the recorded row values in the summaries. Third, for *structured conflict isolation* ($W_2 \rightarrow M_1, M_2$), true cell collisions are kept in VSA choice nodes [43] in catalog metadata rather than triggering abort storms.

Finally, beyond standard lakehouse pipelines, we evaluate SemLake on high-concurrency autonomous agent swarms (W_{swarm}) [4, 20, 28, 41]. When N AI agents concurrently fork a shared table, run a mix of W_1, W_2, and W_3 operations, and race to advance `main` via catalog compare-and-swap (CAS), SemLake turns lost CAS races into server-side semantic merges with group commit, evaluated in simulation for M_1–M_3 under heavy branch fan-in.

In summary, this paper makes five contributions:

(1) **Unified Three-Way Merge Model:** We formalize four criteria for W_1–W_4 (disjoint convergence on W_1, VSA hotspot isolation on W_2, purge-safe merging on W_3, and schema subtyping on W_4) and prove that purge-safety and two-branch serializability (B_6) are incomparable (Section 3).
(2) **Two-Tier Predicate Verifier:** A metadata-time static overlap analyzer paired with a check of recorded row values that eliminates false conflicts without reading data files (Section 4).
(3) **DAG Replay and Bounds (M_1–M_4):** An exact replay-since-fork algorithm over diamond commit DAGs with Roaring DV reconciliation, backed by formal correctness (M_1), latency (M_2), S3 API cost (M_3), and post-merge scan overhead (M_4) bounds (Sections 4, 5).
(4) **Resource-Conscious Prototype:** We implement SemLake over Parquet, Roaring bitmaps, DuckDB, and SQLGlot, paired with an S3 cost/latency store, a live 1,000-file S3 run, native Dolt/`delta-rs` checks, and a swarm simulator (Section 6).
(5) **Experimental Evaluation:** We evaluate SemLake across W_1–W_4, 77 TPC-H refresh pairs, SF 1/SF 10 scans, and W_{swarm} against B_1–B_6 along M_1–M_4 (Section 7).

2 BACKGROUND AND MOTIVATION

2.1 Lakehouse Storage, Deletion Vectors, and Cloud Costs

Open lakehouse table formats such as Delta Lake [6], Apache Iceberg [5], and Apache Hudi [23, 35] layer transactional metadata over immutable columnar Parquet files on cloud object stores [21, 40]. A table state is defined by an immutable snapshot pointing to a manifest tree of data files. Because S3 objects cannot be modified in place, early lakehouse deployments implemented row-level `UPDATE` and `DELETE` operations via copy-on-write (CoW): modifying a single record inside a 512 MB Parquet file required reading the entire file and writing a new 512 MB copy [10], resulting in extreme write amplification (WAF $\gg 1$).

As Durner et al. [15] demonstrate, cloud object storage performance and billing for metadata-intensive operations are governed by API request latencies (30 ms to 100 ms per call) and per-request pricing ($0.005 per 1,000 `PUT`/`LIST` calls and $0.0004 per 1,000 `GET` calls) rather than raw bandwidth. To avoid full-file rewrites, Okolnychyi et al. [31], the Apache Iceberg v3 specification [5], and Delta Lake deletion vectors [27] adopt merge-on-read (MoR) with positional deletion vectors (DVs). A DV is a compressed Roaring bitmap associated with a base Parquet file that marks the zero-based positions of deleted or superseded rows. An update sets the old row position in the base file's DV and appends the updated record

to a small delta Parquet file. Analytical readers apply the bitmap mask during scans, and background compaction periodically folds DVs back into clean base files. Iceberg v3 further assigns stable row lineage identifiers so logical row identity is preserved across updates.

2.2 Problem Taxonomy (W_1–W_4) and Design Metrics (M_1–M_4)

While LST-Bench [10] benchmarks single-branch table operations, multi-branch development adds the *Branch-Execute-Merge* lifecycle $T_0 \xrightarrow{\text{Branch}} (T_A, T_B) \xrightarrow{\text{Execute}} (T'_A, T'_B) \xrightarrow{\text{Merge}} T_M$. Merging T_A and T_B relative to their lowest common ancestor T_0 gives rise to the workload classes W_1–W_4 of Section 1, which we score as follows. M_1 is recall and precision on W_3, W_4 [12, 34], cell-collision isolation on W_2, and false-abort rate (FAR) on W_1 (target 0%). M_2 is wall-clock three-way diff and commit latency across manifest depth, DAG history depth, and contention density $\rho \in (0, 1]$ [23, 30]. M_3 is S3 API requests, dollar cost under the Durner et al. model [15], and write amplification factor (WAF, target ≈ 1). M_4 is the relative latency of DuckDB TPC-H scans on merged snapshots T_M before compaction [10, 37] (target $< 8\%$ for localized updates).

2.3 Why Existing Merge Mechanisms Fail (B_1–B_6 and B_1R)

Dataset versioning systems [7, 8, 22, 29, 36, 39] branch without semantic merge checks, and program-merge verifiers [12, 13, 34, 42, 43] work on kilobyte-scale ASTs. We group existing mechanisms into baseline families B_1–B_6 (plus B_1R); Table 1 summarizes them:

- B_1**: File-level OCC (Iceberg/Delta commits)** [6] intersects modified file paths since T_0: false aborts on W_1, abort storms on W_2, and no read tracking for W_3.
- B_1**R: Delta row-concurrency variant** [27] resolves W_1 writes via DVs but tracks reads at file granularity (`ConcurrentDeleteReadException`), blocking a query when any row in a file it scanned is deleted.
- B_2**: Copy-on-write row merge (Bauplan, lakeFS)** [36, 38] resolves row IDs with no read penalty but rewrites every touched file (WAF $\gg 1$).
- B_3**,** B_4**: Tuple versioning and CRDT views** [2, 17, 22, 29] rebuild per-file tuple dictionaries ($O(N_{\text{files}})$ requests) or converge by last-writer-wins, and neither checks reads.
- B_5**: Dolt** [14] merges Prolly trees per cell and validates declared `FOREIGN KEY` and `CHECK` constraints, but cannot express a query's read predicate and rejects a `CHECK` that legacy T_0 rows violate.
- B_6**: SSI** [1, 9, 16] admits the one-way schedule $T_B \prec T_A$, which purge-safety rejects (Proposition 3.6).

2.4 Extension Beyond Lakehouse Pipelines: Concurrent Agent Workloads (W_{swarm})

While three-way semantic merge is primarily designed for lakehouse ETL, backfill, and WAP pipelines (W_1–W_4), emerging autonomous agent workflows (e.g., Git4Data [20], Chronos [41], BranchBench [4], ATR [28]) extend database branching to multi-agent swarms. We define W_{swarm} as a composite fan-in workload where $N \in [8, 64]$ autonomous agents concurrently fork from `main`, execute mixed SQL operations combining W_1 (disjoint row updates), W_2 (hotspot key updates), and W_3 (range deletes vs. aggregate queries), and race to publish via catalog CAS. A server-side merge service with group commit (Section 4.7) turns lost CAS races into metadata-time semantic merges.

3 FORMAL DATA MODEL AND THREE-WAY MERGE INVARIANTS

3.1 Snapshot DAG, Commit Summaries, and Branch Deltas

Let $\mathcal{R}$ be a universe of stable row identifiers (matching Iceberg v3 row lineage IDs [5]) and $\mathcal{A}$ be a universe of column names.

Definition 3.1 (Table Snapshot). A table snapshot is a tuple $T = (\text{id}, P, S, \rho, \Phi)$, where id is a unique snapshot identifier, $P \subseteq \text{id}^*$ is a tuple of parent snapshot identifiers ($|P| \in \{0, 1, 2\}$), $S = (a_1 : \tau_1, \ldots, a_k : \tau_k)$ is the table schema, $\rho : \mathcal{R} \rightharpoonup (\mathcal{A} \rightharpoonup \mathcal{V})$ is a partial map from live row IDs to column-value records, and Φ is a map from file identifiers to physical file descriptors (k_{obj}, size, token, DV).

A snapshot with $|P| = 1$ is a regular commit; a snapshot with $|P| = 2$ is a merge commit. The snapshots of a repository form a directed acyclic graph (DAG) $G = (V, E)$ with edges (u, p) for each $p \in P(u)$. Each snapshot u has a generation number $\text{gen}(u)$, defined as 0 if $P(u) = \emptyset$ and $1 + \max_{p \in P(u)} \text{gen}(p)$ otherwise. Let $\text{anc}(u)$ denote the set of all ancestors of u (including u itself).

Definition 3.2 (Fork Point). Given two branch tips T_A and T_B, their fork point is the shared ancestor of maximum generation: $T_0 = \text{LCA}(T_A, T_B) = \arg\max_{u \in \text{anc}(T_A) \cap \text{anc}(T_B)} \text{gen}(u)$, breaking ties deterministically by snapshot identifier.

Definition 3.3 (Commit Summary and Folded Branch Delta). Each non-merge commit c records a delta summary $\delta(c) = (\text{ops}_c, S_c, \Pi_c, \sigma_c)$, where $\text{ops}_c : \mathcal{R} \rightharpoonup \{\texttt{insert}, \texttt{update}, \texttt{delete}\} \times (\mathcal{A} \rightharpoonup \mathcal{V})$ maps modified row IDs to operation kinds and column values; S_c is the schema; Π_c is a tuple of attached semantic predicates; and $\sigma_c : \mathcal{R} \rightharpoonup (\mathcal{A} \rightharpoonup \mathcal{V})$ maps deleted rows to their pre-images immediately before deletion. For branch tip T_B and fork T_0, let $C(T_0, T_B) = \{c \in \text{anc}(T_B) \setminus \text{anc}(T_0)\}$. Folding $C(T_0, T_B)$ in topological order yields the folded branch delta $\Delta_B = (W(B), D(B), S_B, \Pi(B))$, comprising write set $W(B) = (\text{ops}_B, \sigma_B)$, deleted row set $D(B) = \{r \mid \text{ops}_B(r) = (\texttt{delete}, \cdot)\}$, tip schema S_B, and predicates $\Pi(B)$.

3.2 Unified Three-Way Merge Criteria Across W_1–W_4

A valid merged snapshot $T_M = \text{merge}(T_0, T_A, T_B)$ must simultaneously satisfy four formal criteria corresponding one-to-one with W_1–W_4:

Criterion 1 (W_1: Disjoint Row and Column Convergence). For each row ID $r \in \mathcal{R}$ modified in $\text{ops}_A \cup \text{ops}_B$: if r is modified by only one branch $X \in \{A, B\}$ or modified identically by both ($\text{ops}_A(r) = \text{ops}_B(r)$), $\rho_M(r)$ adopts $\text{ops}_X(r)$ directly; if both branches update r on disjoint column subsets ($\text{cols}_A(r) \cap \text{cols}_B(r) = \emptyset$), the updates commute and converge into $\rho_M(r) = \rho_0(r) \oplus \Delta_A(r) \oplus \Delta_B(r)$ regardless of physical file co-location.

Table 1: Architectural comparison of existing merge and concurrency paradigms (B_1–B_6 and B_1R) against SemLake across all four lakehouse workload classes (W_1–W_4) and evaluation metrics (M_1–M_4). Baseline cells describe each system's documented rule; Section 6 states which baselines are measured systems.

System / Baseline	Granularity	W_1: Shared-File Updates (M_1, M_3, M_4)	W_2: Hotspot Key Collisions (M_1, M_2)	W_3: Semantic Dependencies (M_1)	W_4: Schema & Manifest Scale (M_1, M_2, M_3)
B_1: File-level OCC [6]	Physical file	Aborts (FAR = 100% on shared files)	Whole-branch file-conflict abort storm	Commits silently (0% recall)	$O(N_{\text{files}})$ manifest scan; no 3-way schema unify
B_1R: Delta row-write variant of B_1 [27]	Row write DV + file read	Commits via DVs (FAR = 0%, low WAF)	Row-conflict abort & full transaction retry	False-blocks all readers of touched file	$O(N_{\text{files}})$ manifest scan on merge
B_2: Copy-on-Write row merge [36]	Row ID (CoW)	Full file rewrite (WAF $\gg$ 1, 0% scan pen.)	Whole-branch abort & full file rewrite	Commits silently (0% recall)	O(TableBytes) full file rewrite I/O
B_3: Relational tuple versioning [22, 29]	Tuple version dictionary	Rebuilds tuple-version records for every touched file	Whole-branch primary-key conflict abort	Commits silently (0% recall on reads/domains)	$O(N_{\text{files}})$ S3 API calls to rebuild tuple dictionaries
B_4: Automatic CRDT views (CRDV) [17]	Operation log	Commits via log	Silent LWW overwrite (loses updates)	Commits silently (0% recall)	$O(\lvert\text{Log}\rvert)$ replay; static view schema
B_5: Constraint-aware cell merge (Dolt) [14]	Prolly-tree cell + SQL constraint	Commits via cell diff	Halts merge in dirty cell-conflict state	Misses query reads (recall 0.11; FK/CHECK only)	$O(\log N_{\text{cells}})$ tree diff; rejects legacy-violating CHECK
B_6: Serializable Snapshot Isolation (SSI) [9]	Online OLTP SIREAD lock	Commits if tuple locks disjoint	Interactive transaction abort on ww-conflict	Permits 1-way $T_B \prec T_A$ (violates purge-safety)	Volatile lock table; no branched S3 manifests
SemLake (Ours)	**Row DV + SQL Predicate + VSA**	**Roaring DV union (FAR = 0%, no file rewrite, +0.9% localized scan overhead)**	**Composes disjoint cols; isolates cell collisions in VSA ($\mathcal{V}_M$)**	**Two-tier predicate & pre-image check (100% recall)**	**Bidirectional schema unify + $O(\Delta_{\text{manifest}})$ delta manifest (7 S3 calls)**

Table 2: Taxonomy of predicate kinds supported by SemLake for workload W_3, their SQL or programmatic origin, and the formal three-way conflict rule evaluated between branch deltas Δ_A and Δ_B.

Predicate Kind	Attached By	Representative Example	Three-Way Conflict Rule (Δ_A vs. Π_B)
reads_where(φ_R)	Reader / aggregate commit (W_3)	kind = 'vip' AND value >= 10	Conflicts with deletes_where(φ_D) $\in \Pi(A)$ if Overlap(φ_D, φ_R) holds and the deleted-row check does not clear it, and with an update or insert in ops$_A$ that moves a row across φ_R or changes a column the reader uses on a row inside it.
deletes_where(φ_D)	Delete / purge commit (W_3)	l_shipdate < 19930201	Declares the selection scope of a delete operation; paired symmetrically against every reads_where(φ_R) $\in \Pi(B)$.
rows_live(I)	Point-read / lineage commit (W_3)	$I = \{101, 102, 105\}$	Conflicts if the partner branch deletes any listed row identifier: $D(A) \cap I \neq \emptyset$.
foreign_key(I)	Child-row insert commit (W_3)	$I = \{1\}$ (orders.user_id → users.id)	Conflicts if the partner branch deletes any referenced parent row identifier: $D(A) \cap I \neq \emptyset$.
domain(φ_{dom})	Invariant / policy commit (W_3)	value >= 0	Conflicts if any insert or update in ops$_A$ assigns column values that violate φ_{dom}.

Criterion 2 (W_2: Hotspot Cell-Collision Isolation via VSA). If both branches assign distinct values $v_A \neq v_B$ to the same column $c \in \text{cols}_A(r) \cap \text{cols}_B(r)$ or one updates r while the other deletes r, the conflicting cell is encapsulated without data loss into a Version Space Algebra (VSA) [43] choice node choice$(r, c, [(A, v_A), (B, v_B)]) \in \mathcal{V}_M$, while non-conflicting columns of r still converge in $\rho_M(r)$.

Criterion 3 (W_3: Cross-Branch Predicate Preservation). Table 2 defines the five predicate kinds in SemLake: reads_where(φ_R) and deletes_where(φ_D) capture the selection scope of SQL SELECT and DELETE statements; rows_live(I) and foreign_key(I) specify finite row-ID sets $I \subset \mathcal{R}$ that must remain live in T_M; and domain(φ_{dom}) asserts a value invariant over concurrent writes ops$_X$. Unlike a table-wide SQL CHECK constraint in B_5 (Dolt), a domain predicate validates only the new assignments in ops$_X$ since T_0 and does not fail if untouched legacy records in T_0 lie outside φ_{dom}.

Criterion 4 (W_4: Bidirectional Schema Subtyping Compatibility). Let $\preceq_{\text{type}}$ denote the lossless numeric widening preorder (int32 $\prec_{\text{type}}$ int64 $\prec_{\text{type}}$ double, and $\tau \preceq_{\text{type}} \tau$ for all types τ). Tip schemas S_A and S_B are merge-compatible relative to S_0 iff for every column $a \in \mathcal{A}$: (a) neither branch drops or renames a column present in S_0; (b) any column $a \notin S_0$ added by S_A or S_B is nullable (and if added by both, has a common supertype); and (c) for every $a \in S_0$, $\tau_0(a) \preceq_{\text{type}} \tau_A(a)$ and $\tau_0(a) \preceq_{\text{type}} \tau_B(a)$, yielding the unified schema $S_M(a) = \sup_{\preceq_{\text{type}}}(\tau_A(a), \tau_B(a))$ without rewriting base Parquet files.

Definition 3.4 (Three-Way Table Merge Operator). A three-way merge operator merge(T_0, T_A, T_B) evaluates Criteria 1–4 across W_1–W_4: (1) If Criteria 3 and 4 hold, it returns merged snapshot T_M with parents (T_A, T_B), converged row state ρ_M (W_1, status clean, outcome keep when $\mathcal{V}_M = \emptyset$), VSA choices $\mathcal{V}_M$ (W_2, status vsa, outcome replan when $\mathcal{V}_M \neq \emptyset$), and unified schema S_M (W_4); (2) semantic_conflict(V) (block) if predicate violations $V \neq \emptyset$ violate Criterion 3 (W_3); or (3) schema_conflict (block) if schemas violate Criterion 4 (W_4).

3.3 Purge-Safe Merge Invariant vs. Classical Serializability (W_3 vs. B_6)

We now formalize Criterion 3 (W_3) as the *purge-safe merge invariant* and compare it with classical transaction serializability (B_6, SSI) [1, 9, 16].

Definition 3.5 (Purge-Safe Merge). Let T_A, T_B be branches with fork T_0. A merge $T_M = \text{merge}(T_0, T_A, T_B)$ is *purge-safe* if, symmetrically for $(X, Y) \in \{(A, B), (B, A)\}$: (1) for every reads_where(φ_R) $\in \Pi(Y)$ and every row $r \in D(X)$, neither the row's value at the fork nor its pre-image at delete time satisfies φ_R: $\rho_0(r) \not\models \varphi_R$ (when r exists in T_0) and $\sigma_X(r) \not\models \varphi_R$; (2) for every rows_live(I) or foreign_key(I) $\in \Pi(Y), D(X) \cap I = \emptyset$; (3) for every domain($\varphi_{\text{dom}}$) $\in \Pi(Y)$ and $(r, v) \in \text{ops}_X, v \models \varphi_{\text{dom}}$; and (4) for every reads_where(φ_R) $\in \Pi(Y)$ and every row r that X updates or inserts without deleting, with value $\rho_X(r)$ at X's tip, r maintains the same membership under φ_R, namely $\rho_0(r) \models \varphi_R \Leftrightarrow \rho_X(r) \models \varphi_R$ (treating an inserted row as outside φ_R at T_0), and whenever r satisfies φ_R, $\rho_0(r)$ and

$\rho_X(r)$ agree on the projected columns $\text{cols}(\varphi_R)$ referenced outside the `WHERE` clause.

Proposition 3.6 (Purge-Safety vs. Serializability (B_6)). *Purge-safety and two-branch conflict serializability are incomparable. There exist branch pairs (T_A, T_B) that admit a serializable execution order $T_B \prec T_A$ under SSI (B_6) [9] yet violate purge-safety, and purge-safe merges that are not serializable.*

Proof. Consider Figure 1. Branch T_B reads rows satisfying φ_R = (`kind='vip'` $\wedge$ `val` ≥ 10) from T_0 and inserts a summary row into a new file. Branch T_A deletes rows satisfying φ_D = (`kind='vip'`) from T_0 and reads nothing from T_B. In Adya's dependency graph [1], there is a single antidependency $T_B \xrightarrow{rw} T_A$ and no edge $T_A \to T_B$. Because a cycle requires two consecutive *rw*-edges [9], the graph is acyclic and the serial schedule $T_B \prec T_A$ is accepted by SSI. However, in T_M, both the erasure of VIP records and the derived summary coexist in the same published snapshot. Because the deleted VIP rows satisfy φ_R, Definition 3.5 rejects the merge. Conversely, let both branches update the same cell of the same row to different values, with no predicates. Definition 3.5 holds trivially, but any serial order loses one of the two writes, so a merge that keeps both outcomes as a VSA choice (Criterion 2) is not equivalent to a serial execution. Hence neither property implies the other. □

In a branched lakehouse, T_M is published as a single dataset snapshot: if T_A is a regulatory erasure or a scrub of corrupted records, admitting T_B's pre-purge aggregate publishes statistics derived from records that T_M claims were removed. Proposition 3.6 shows that purge-safety and conflict serializability capture different consistency contracts. When an application prefers one-way serializability (accepting $T_B \prec T_A$), SemLake provides a directional mode; by default it enforces symmetric purge-safety.

3.4 Decidable SQL Predicate Fragment ($\mathcal{F}$)

To evaluate $\varphi_D \wedge \varphi_R$ in metadata time without touching data files, SemLake defines a static conjunctive fragment $\mathcal{F}$ over column names $c \in \mathcal{A}$ and literals ℓ: $\varphi ::= t_1 \wedge \cdots \wedge t_m$, where each atomic constraint t_i takes one of three forms: (1) binary comparisons c op ℓ with op $\in \{=, \neq, <>, <, \leq, >, \geq\}$; (2) finite set membership c `IN` $(\ell_1, \ldots, \ell_k)$; or (3) closed intervals c `BETWEEN` ℓ_{low} `AND` ℓ_{high}. Clauses containing `OR`, `NOT`, arithmetic expressions, joins, or subqueries lie outside $\mathcal{F}$ and are conservatively routed to the delete-time row-value post-check (Section 4.4).

4 SYSTEM ARCHITECTURE AND MERGE DESIGN

4.1 Architectural Overview

Figure 2 shows the four components of SemLake: a catalog that keeps branch heads (updated by atomic compare-and-swap) and the snapshot DAG with cached ancestor generations for LCA search; a commit-summary layer that extracts read and delete predicates from SQL and writes one compact summary per commit; the three-way merge engine (replay since the fork, the two-tier conflict check, schema unification, and row reconciliation into converged records or VSA choices); and a storage substrate that keeps base Parquet files immutable and writes Roaring DVs [31], one delta Parquet file, and a delta manifest per merge.

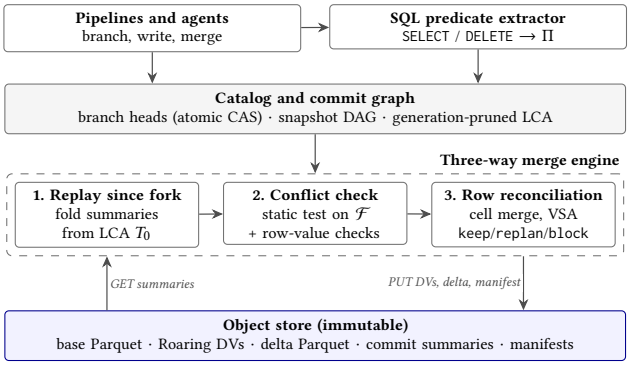


Figure 2: System architecture of SemLake. Merging two diverged branches reads only the compact commit summaries since the fork point T_0 and publishes a single delta Parquet file, updated Roaring deletion vectors, and a delta manifest list.

4.2 Compact Commit Summaries and SQL Extraction

In standard Iceberg or Delta tables, discovering which rows changed between two snapshots requires reading manifest trees and scanning data files or change-data-capture logs. SemLake eliminates data scans during merge by writing one compact JSON summary per commit containing: (1) parent snapshot IDs, branch name, and schema fields (W_4); (2) modified file IDs and their updated descriptors, including base64-encoded Roaring DVs (W_1); (3) row-level operations mapping each modified row ID to its operation kind, column assignments, and the row's values just before the update or delete (W_1, W_2, W_3); (4) attached predicates (W_3); and (5) any open VSA choice nodes (W_2). Replaying a non-merge commit during a three-way merge issues exactly one object-store `GET` request for its summary object, regardless of whether the table contains 10^3 or 10^5 unchanged Parquet files.

Writers can attach predicates programmatically or pass raw SQL statements to SemLake's SQL extractor. The extractor parses `SELECT` and `DELETE` statements into an AST, normalizes the `WHERE` clause into canonical SQL text, and emits `reads_where`(φ_R) or `deletes_where`(φ_D). Queries containing joins, `UNION`, or subqueries retain their full clause text so the conflict verifier conservatively classifies them as outside $\mathcal{F}$ and falls back to the deleted-row check. A read is checked only if it is declared: a pipeline routes its queries through the extractor (for example from a dbt or Spark query hook, which we have not built) or attaches predicates itself, and a reader that cannot declare a clause can attach a table-wide `reads_where`, which any delete then conflicts with.

4.3 DAG-Aware Delta Replay Since the Fork

When long-running branches merge repeatedly across multi-stage pipelines (W_4), the commit history forms a DAG with nested merge diamonds rather than a simple tree. A naive three-way merge that inspects only the tip summary of each parent fails in two ways: first, a merge commit's summary does not duplicate ancestor predicates

Algorithm 1: DAG-Aware Replay Since the Fork (`ReplaySinceFork`)

Input: Repository *repo*, object store *store*, fork ID b, branch tip ID t
Output: Folded branch delta $\Delta_t = (ops, changed, S, \Pi, \Phi, vsa, \sigma)$
1: **if** $t = b$ **then return** EmptyDelta(S_b)
2: *contained* ← *repo*.ancestors(b) ▷ Cached ancestor set of fork b
3: *order* ← TopologicalWalk(t, skip = *contained*) ▷ Post-order DFS
4: *folded*[b] ← EmptyDelta(S_b)
5: **for each** $u \in$ *order* **do**
6: *snap* ← *repo*.snapshots[u]
7: **if** |*snap*.parents| ≥ 2 **then** ▷ Internal merge diamond
8: (p_1, p_2) ← *snap*.parents; f ← LCA(p_1, p_2)
9: **if** $f \notin$ *folded* **then** $f \leftarrow b$
10: L ← StripPrefix(Known(p_1), *folded*[f])
11: R ← StripPrefix(Known(p_2), *folded*[f]); br ← Combine(L, R, S_b)
12: *folded*[u] ← FoldCommit(*folded*[f], br, ρ_b)
13: **else** ▷ Regular commit: 1 summary `GET`
14: p ← *snap*.parents[0] **if** *snap*.parents **else** b
15: *step* ← ParseSummary(*store*.get(*snap*.summary_key))
16: *folded*[u] ← FoldCommit(Known(p), *step*, ρ_b)
17: **return** *folded*[t]

Algorithm 2: Two-Tier Semantic Conflict Check (`VerifyMergeSemantics`)

Input: Folded branch deltas Δ_A, Δ_B, fork rows ρ_0
Output: Violation list V ($V = \emptyset$ iff merge is purge-safe)
1: $V \leftarrow [\,]$; D_X, U_X ← deleted, changed IDs in Δ_X.ops, $X \in \{A, B\}$
2: **for each** $(X, Y) \in \{(A, B), (B, A)\}$ **do**
3: $moved_X \leftarrow \{r \in D_X \mid \sigma_X(r) \neq \rho_0(r)\}$ ▷ Rows updated before deletion
4: **for each** $p \in \Delta_Y.\Pi$ **with** p.kind ∈ {`rows_live`, `foreign_key`} **do**
5: **if** p.row_ids ∩ $D_X \neq \emptyset$ **then** Append(V, Violation(p, p.row_ids ∩ D_X))
6: **for each** $s \in \Delta_X.\Pi$ **with** s.kind = `deletes_where` **do**
7: **for each** $p \in \Delta_Y.\Pi$ **with** p.kind = `reads_where` **do**
8: **if not** WhereOverlap($s.\varphi_D, p.\varphi_R$) **then** $R \leftarrow moved_X$; $vals \leftarrow \{\rho_0\}$
9: **else** $R \leftarrow D_X$; $vals \leftarrow \{\rho_0, \sigma_X\}$
10: **if** AnyMatch($R, p.\varphi_R, vals$) **then** Append(V, Violation(p, s))
11: **for each** $p \in \Delta_Y.\Pi$ **with** p.kind = `domain` **do**
12: $bad \leftarrow \{r \mid (r, (op, vals, \cdot)) \in \Delta_X.\text{ops},\ op \in \{\texttt{insert}, \texttt{update}\},\ vals \not\models p.\varphi_{\text{dom}}\}$
13: **if** $bad \neq \emptyset$ **then** Append(V, Violation(p, bad))
14: **for each** `reads_where` $p \in \Delta_Y.\Pi$ **with** $M \leftarrow \{r \in U_X \mid \text{Unstable}(r, p)\} \neq \emptyset$ **do** Append(V, Violation(p, M))
15: **return** V

or rows parked in VSA choices, so a subsequent merge would forget earlier read or delete constraints (W_3); second, walking all ancestors of an internal merge commit without stripping commits already present in the new fork point $b = \text{LCA}(T_A, T_B)$ would either replay shared history prefixes as false conflicts (W_1, W_2) or drop commits whose parent predates b.

Algorithm 1 (`ReplaySinceFork`) reconstructs the exact folded delta Δ_t of branch tip t since fork point b. First, it retrieves the cached ancestor set *contained* = anc(b) and executes a post-order depth-first walk from t that halts upon entering *contained*. Every commit in anc(t) \ anc(b) is emitted once in topological order (parents before children), mapping any parent inside anc(b) to *folded*[b] via Known(p).

Second, for each regular commit u, Algorithm 1 fetches u's summary object (1 `GET`) and folds it onto its parent's delta via `FoldCommit`. If u updates a row r previously parked in an open VSA choice, `FoldCommit` resolves the VSA choice for r and records the new write; if u deletes r, it records r's delete-time pre-image in $\sigma[r]$. Third, when the walk encounters an internal merge node u with parents (p_1, p_2), it locates u's internal fork $f = \text{LCA}(p_1, p_2)$, strips the shared prefix *folded*[f] from both parent deltas via `StripPrefix`, and combines their relative deltas without extra object-store reads.

4.4 Two-Tier Semantic Conflict Verification (W_3)

Given branch deltas Δ_A, Δ_B, Algorithm 2 (`VerifyMergeSemantics`) evaluates Criterion 3 (Definition 3.5) symmetrically for $(X, Y) \in \{(A, B), (B, A)\}$ across four verification stages:

1. Tier-1 Static Satisfiability (`WhereOverlap`*).* For each pair of `deletes_where`(φ_D) ∈ $\Delta_X.\Pi$ and `reads_where`(φ_R) ∈ $\Delta_Y.\Pi$, `Where Overlap` tests whether $\varphi_D \wedge \varphi_R$ is satisfiable. When both clauses parse into the conjunctive fragment $\mathcal{F}$, atomic constraints are grouped by column c. For each column c, the solver intersects the = and `IN` candidates, removes ≠ points, and keeps the tightest lower and upper bounds with their strictness; c is unsatisfiable iff the interval is empty or excludes every remaining candidate. If any column is unsatisfiable, $\varphi_D \wedge \varphi_R$ is unsatisfiable and `Where Overlap` returns `False` in microseconds without reading a single table row. No row deleted under φ_D can then satisfy φ_R with its delete-time values; the one remaining case is a row that X updated before deleting it, whose fork value (which Y read) may still satisfy φ_R, so Line 8 checks the fork values of exactly those rows.

2. Tier-2 Deleted-Row Value Check (`AnyMatch`*).* When the static test cannot rule out an overlap, either because a clause falls outside $\mathcal{F}$ (e.g., `DELETE WHERE NOT kind = 'vip'`) or because the two clauses overlap on a value interval containing no deleted row, Lines 9–10 evaluate φ_R on each deleted row's recorded value at the fork (ρ_0) and at delete time (σ_X). Both come from the commit summaries, so no base Parquet file is read. Clauses in $\mathcal{F}$ are evaluated natively, and other single-table SQL clauses are evaluated by DuckDB [33] over those rows. A row without a recorded pre-image, or a read clause with a join or subquery, conservatively retains the conflict. This step costs $O(|D_X|)$ per clause pair.

3. Lineage, Foreign-Key, and Domain Verification. Lines 4–5 intersect each `rows_live` and `foreign_key` set $I \in \Delta_Y.\Pi$ against D_X in $O(\min(|I|, |D_X|))$ time. Lines 11–13 evaluate each `domain`(φ_{dom}) ∈ $\Delta_Y.\Pi$ against all `insert` and `update` assignments in Δ_X.ops, flagging any record whose assigned columns violate φ_{dom}.

4. Read Stability under Updates and Inserts (`Unstable`*).* Line 14 checks Condition 4 on each updated or inserted row's fork and tip values from the summaries using the evaluator of `AnyMatch`: `Unstable` flags a row whose membership under φ_R changes between $\rho_0(r)$ and $\rho_X(r)$, or a row matching φ_R in both states whose modified columns intersect cols(φ_R), the columns referenced outside the `WHERE` clause (`count(*)` uses none; `SELECT *` or a join uses all). Any modified row on which φ_R cannot be evaluated is flagged as crossing, in $O(|U_X|)$ time per read predicate.

4.5 Unified State Reconciliation Across W_1–W_4

Combining Algorithm 2 with Criteria 1–4 yields a unified decision and state-reconciliation procedure across W_1–W_4 (where at runtime, the W_3 and W_4 `block` checks execute first as fail-fast metadata guards prior to materializing W_1 and W_2 row deltas):

(1) **Disjoint row/column convergence (W_1 → `keep`):** Under Criterion 1, rows modified by only one branch or identically on both adopt $\text{ops}_X(r)$. Rows updated on disjoint columns commute into $\rho_0(r) \oplus \Delta_A(r) \oplus \Delta_B(r)$ with zero false aborts (M_1).
(2) **Hotspot collision isolation in VSA (W_2 → `replan`):** Under Criterion 2, if both branches assign distinct values $v_A \neq v_B$ to column c (or update vs. delete), SemLake constructs a VSA choice node [43] `choice`$(r, c, [(A, v_A), (B, v_B)]) \in \mathcal{V}_M$. Non-conflicting columns merge cleanly, while the cell retains its base value in the snapshot until resolved (M_1, M_2).

(3) **Semantic conflict blocking ($W_3 \rightarrow$ `block`):** Under Criterion 3, if Algorithm 2 returns $V \neq \emptyset$, the merge halts immediately with `semantic_conflict`(V) before writing any data files (M_1).
(4) **Bidirectional schema unification ($W_4 \rightarrow$ `keep`/`block`):** Under Criterion 4, additive nullable columns and numeric widenings (`int32` $\rightarrow$ `int64` $\rightarrow$ `double`) unify into S_M without rewriting base files; incompatible column drops, renames, or narrowings halt with `schema_conflict` (`block`).

4.6 Zero-Copy Physical Merge over Deletion Vectors (M_3, M_4)

SemLake materializes the merged snapshot T_M without rewriting any base Parquet file (M_3, M_4). For every row r updated or deleted in ops_M that resided in a base file f at zero-based offset $\text{pos}_0[r]$, SemLake sets bit $\text{pos}_0[r]$ in f's Roaring deletion vector ($\text{DV}_M(f) = \text{DV}_A(f) \cup \text{DV}_B(f) \cup \{\text{pos}_0[r]\}$). All newly inserted records and resolved row updates across both branches are serialized into a single compact delta Parquet file (`data/<id>/delta-<id>.parquet`).

The commit protocol then writes three small metadata objects: (1) the merge summary `manifests/<id>.summary.json`, (2) a delta manifest `manifests/<id>.delta.json` containing descriptors only for files whose DV or payload changed since T_0, and (3) a two-element manifest list `[base_manifest_key,delta_key]`. Because untouched base file descriptors are inherited by reference, S3 API requests scale as $O(\Delta_{\text{manifest}})$ (M_3). During subsequent analytical queries (M_4), the scan engine inspects each row group's DV: clean row groups ($\text{DV} = \emptyset$) execute native vectorized Parquet scans without filtering overhead, while background compaction periodically folds DVs back into base files once the dirty-row ratio or scan-to-merge ratio crosses the cost threshold.

4.7 Extension: Commit Service for Agent Swarms (W_{swarm})

To support the composite agent-swarm workload W_{swarm} beyond standard WAP pipelines, SemLake provides a server-side CAS merge service (`swarm_service.py`). When k agents concurrently invoke CAS(`main`, T_{expected}, T_{local}), the $k-1$ losing branches are not aborted; instead, the server executes a three-way semantic merge between each losing snapshot and the advancing head (M_1), rejecting only branches that conflict with commits already on `main`. Under one CAS per commit, throughput is bounded by per-merge catalog and object-store round trips (M_2). Under *group commit*, the service takes up to G queued branches, merges them sequentially onto the head in memory, and publishes the batch with one catalog CAS (evaluated in Section 7.7).

5 CORRECTNESS, COMPLEXITY, AND SCAN OVERHEAD ANALYSIS

5.1 Soundness and Exactness Proofs (M_1)

Theorem 5.1 (Soundness). *Let T_A and T_B be two branches with fork point T_0, and suppose every row deleted on a branch is deleted by a statement whose* `deletes_where` *clause the row satisfied at delete time, and that the commit summaries record each deleted row's pre-image. If Algorithm 2 returns $V = \emptyset$ for the deltas produced by Algorithm 1 and SemLake commits $T_M = \text{merge}(T_0, T_A, T_B)$, then T_M is purge-safe (Definition 3.5) and satisfies Criteria 1–4 across W_1–W_4.*

Proof. We verify Conditions 1–4 of Definition 3.5. For Condition 2, Lines 4–5 of Algorithm 2 test p.row_ids $\cap D_X \neq \emptyset$ for all `rows_live` and `foreign_key` predicates on both branches; $V = \emptyset$ guarantees $D_A \cap I = \emptyset$ and $D_B \cap I = \emptyset$. For Condition 3, Lines 11–13 test every conjunct of each `domain`(φ_{dom}) predicate against all `insert` and `update` assignments on the partner branch; $V = \emptyset$ guarantees all assigned column values satisfy φ_{dom}.

For Condition 1, fix `reads_where`(φ_R) $\in \Pi(Y)$ and $r \in D(X)$. By assumption, $\sigma_X(r) \models \varphi_D$ for some `deletes_where`(φ_D) $\in \Pi(X)$. If `WhereOverlap` is true, Lines 9–10 found that neither $\rho_0(r)$ nor $\sigma_X(r)$ satisfies φ_R. Otherwise `WhereOverlap` returned `False`, so some column c admits no value satisfying $T_c(\varphi_D) \wedge T_c(\varphi_R)$ under the per-column test of Section 4.4; hence $\varphi_D \wedge \varphi_R$ is unsatisfiable, so $\sigma_X(r) \not\models \varphi_R$; if $r \in \textit{moved}_X$, Line 10 found $\rho_0(r) \not\models \varphi_R$, and otherwise $\rho_0(r) = \sigma_X(r)$. For Condition 4, Line 14 flags every changed row that crosses φ_R, changes $\text{cols}(\varphi_R)$ inside it, or cannot be evaluated. Criteria 1, 2, and 4 follow directly from the column-disjoint composition, VSA encapsulation, and $\preceq_{\text{type}}$ schema unification rules in Section 4.5. □

Lemma 5.2 (Exact DAG Replay Since the Fork). *For any branch tip t and fork point $b = LCA(t_{left}, t_{right})$, Algorithm 1 folds every non-merge commit $c \in \text{anc}(t) \setminus \text{anc}(b)$ once in topological order, excludes every commit in $\text{anc}(b)$, and preserves all predicates and open VSA choices across intermediate clean merges.*

Proof. The post-order DFS from t halts along any edge entering *contained* and emits each node $u \notin$ *contained* only after all of its parents outside *contained* have been emitted, yielding $\text{anc}(t) \setminus \text{anc}(b)$ in topological order. Halting specifically on $\text{anc}(b)$ avoids dropping valid branch commits in multi-merge chains: if a report branch R (forked from T_0) merges with an update branch U to form M_1, and M_1 later merges with a purge branch P that forked from U, the new LCA is $b = U$. Even though R is not a descendant of U, $R \in \text{anc}(M_1) \setminus \text{anc}(U)$ is retained and its parent $T_0 \in \text{anc}(U)$ maps to $\textit{folded}[b]$ via Known(p), preserving R's read predicate. Meanwhile, for any internal merge node u with fork f, `StripPrefix` removes $\textit{folded}[f]$ before combining its parents, so shared history prefixes are never double-counted as concurrent conflicts. □

5.2 Precision, Complexity, and Scan Overhead Bounds (M_1–M_4)

Precision Boundaries (M_1). Theorem 5.1 and Lemma 5.2 rule out false negatives on W_3, W_4 and false aborts on W_1. The check can still conservatively block a safe merge when: (a) the read clause contains a join or subquery that cannot be evaluated on single-table rows; (b) a row entered φ_R only on branch X before being deleted on X, which Definition 3.5 counts as a match; or (c) an update changes a column in $\text{cols}(\varphi_R)$ without changing the query result. We cross-check Theorem 5.1 on 200 randomized branch pairs, merged directly and across intermediate merges in both orders against a brute-force oracle, observing zero missed conflicts.

Merge Time and Cloud S3 API Complexity (M_2, M_3). Let $K = |C(T_0, T_A)| + |C(T_0, T_B)|$ be the number of non-merge commits since

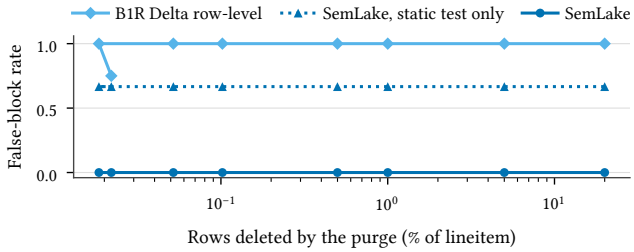


Figure 3: False-block rate vs. delete selectivity ($W_3 \rightarrow M_1$) on TPC-H `lineitem` (SF 0.01, 31 files): a purge whose clause matches none of the rows a concurrent Q6-style report reads (6 of the 31 files). B_1R's file-level read check blocks 0.75 of the merges at the smallest purge and all of them from 0.02% upward. The static test alone blocks the two clause forms outside $\mathcal{F}$ (`NOT`, `OR`; rate 0.67); with the deleted-row check SemLake blocks none.

T_0, $|\Delta_{\text{rows}}|$ be the number of modified rows, $|\Delta_{\text{files}}|$ be the number of touched files, and $|\Pi|$ be the number of attached predicates. Algorithm 1 issues at most K object-store `GET` requests (M_3), Algorithm 2 and VSA reconciliation run in $O(|\Pi_A| \cdot |\Pi_B| + |\Delta_{\text{rows}}|)$ CPU time (M_2), and physical publication writes 1 delta Parquet file plus 3 small JSON objects (4 `PUT`s), yielding $O(K + |\Delta_{\text{files}}|)$ total S3 API cost and WAF $= 1 + B_{\text{meta}}/B_{\Delta\text{rows}} \approx 1$ (M_3), independent of total table file count N_{files} and row count N_{rows}.

Post-Merge Analytical Scan Overhead Bound (M_4). Let a table span G_{rg} Parquet row groups, and let $G_{\text{dirty}} \leq G_{\text{rg}}$ be the number of row groups with non-empty Roaring DVs after merging. Because clean row groups (DV $= \emptyset$) execute native vectorized scans at base cost C_{scan} while each dirty row group incurs a relative bitmap-mask factor α, post-merge query latency is bounded by $T_{\text{scan}}(T_M) = C_{\text{scan}} \cdot (1 + \alpha \cdot \frac{G_{\text{dirty}}}{G_{\text{rg}}})$. For localized updates (W_1, where $G_{\text{dirty}} \ll G_{\text{rg}}$), relative scan overhead approaches 0. When cumulative scan overhead across N_{scans} queries exceeds compaction rewrite cost C_{compact} (at crossover point $N^*_{\text{scans}} = \frac{C_{\text{compact}}}{\alpha \cdot C_{\text{scan}} \cdot (G_{\text{dirty}}/G_{\text{rg}})}$), background compaction folds DVs into clean base files and resets $G_{\text{dirty}} \rightarrow 0$.

6 IMPLEMENTATION

Under single-machine academic constraints, we implemented SemLake in Python (about 2,500 lines in eight modules: `metadata.py`, `storage.py`, `merge.py`, `verification.py`, `sql_predicates.py`, `reader.py`, `s3store.py`, and `baselines.py`, plus about 5,800 lines of benchmark and simulation code).

Data files are encoded as Apache Parquet objects via `pyarrow`, with a hidden `__row_id` column tracking Iceberg v3 row lineage [5]. Positional deletion vectors use `pyroaring` (`BitMap.serialize` and bitwise union `|`). Analytical queries in metric M_4 execute in DuckDB over local Parquet files; exported snapshots (`reader.py`) read clean row groups directly and apply non-empty DVs via a positional boolean mask join. To support deep commit histories and multi-branch fan-in in memory, snapshot row states use a copy-on-write structural overlay (`RowView`): each commit stores only modified records and deleted row IDs, walking the parent chain on lookup without copying unchanged rows. In the commit graph, each snapshot caches its ancestor-generation map upon creation, allowing `lca` to run a generation-ordered max-heap search that terminates as soon as the highest open generation falls below the best common ancestor found.

To profile cloud storage costs deterministically (M_3) while verifying physical behavior, our in-memory object store (`MemoryObjectStore`) and live S3 backend (`s3store.py`) share the same `put/get/list` interface and log exact request counts and transferred bytes, priced under AWS S3 Standard (`us-east-1`: $0.005 per 1,000 `PUT/LIST` calls, $0.0004 per 1,000 `GET` calls [3]) with analytical cloud latency $T_{\text{est}} = T_{\text{cpu}} + N_{\text{put}} \cdot 100\,\text{ms} + N_{\text{get}} \cdot 30\,\text{ms} + N_{\text{list}} \cdot 50\,\text{ms} + B_{\text{put}}/(100\,\text{MB/s})$. All reported CPU latencies are real wall-clock timings without synthetic sleeps, and Section 7.4 calibrates the model against a live S3 bucket in `eu-north-1`. B_1, B_1R, B_2, and B_6 are rule models evaluated on the same snapshots as SemLake (with B_1 cross-checked against a real `delta-rs` table and a real lakeFS server), B_3 and B_4 are storage-layout reimplementations, and B_5 invokes the native Dolt binary via `dolt merge`.

7 EXPERIMENTAL EVALUATION

7.1 Experimental Setup and Baselines

All experiments execute on one machine with 4 cores, Python 3.13.16, DuckDB 1.5.6, `pyarrow` 26.0.0, `pyroaring`, `sqlglot`, Dolt, and a lakeFS [38] 1.88 server; the result-oracle, pre-image, lakeFS and second S3 runs used Python 3.13.17 and DuckDB 2.0.0. TPC-H data is real `lineitem` generated with `tpchgen-cli`, a port of the TPC-H generator [37]. Timed merges run once untimed and then 5 times, and we report the median with the range; DuckDB scans run in separate processes; the swarm runs 5 seeds and reports means with 95% confidence intervals. Following the Branch-Execute-Merge protocol adapted from LST-Bench [10], we evaluate all workloads in canonical order: W_1 (shared-file disjoint updates), W_2 (hotspot key contention $\rho \in [0\%, 20\%]$), W_3 (cross-branch predicate conflicts over a 34-case labelled semantic suite spanning W_1, W_3, W_4 and 42 TPC-H [37] refresh-versus-query branch pairs), W_4 (schema evolution and manifest/DAG scaling from 1K to 10K files and 42 to 360 snapshots), and the extension workload W_{swarm} (8 to 64 concurrent agents combining W_1–W_3).

Following Table 1, we compare SemLake against all six baseline families in canonical order (B_1, B_1R, B_2, B_3, B_4, B_5, B_6): B_1 (File-level OCC [6]), B_1**R** (Delta row-write variant of B_1 with file-level read conflict checks [27]), B_2 (Copy-on-Write row merge [36]), B_3 (Relational tuple versioning [22, 29]), B_4 (Automatic CRDT views [17]), B_5 (Constraint-aware cell merge, the Dolt binary executing native `dolt merge` with SQL `FOREIGN KEY` and `CHECK` constraints [14]), and B_6 (Classical SSI [9], detecting two-edge *rw*-antidependency cycles). B_5 and lakeFS are external systems; B_1, B_1R, B_2 and B_6 are rule models and B_3, B_4 our reimplementations (Section 6). lakeFS merges branches object by object; we give it its most favourable layout, one object per data file and per DV and uniquely named metadata objects, so only data-level overlap conflicts. All baselines are scored on M_1 (Table 3); the storage benchmarks (M_2–M_4) cover B_1, B_1R and B_2–B_4, because B_5 keeps its own Prolly-tree store and B_6 uses OLTP `SIREAD` locks without Parquet manifests.

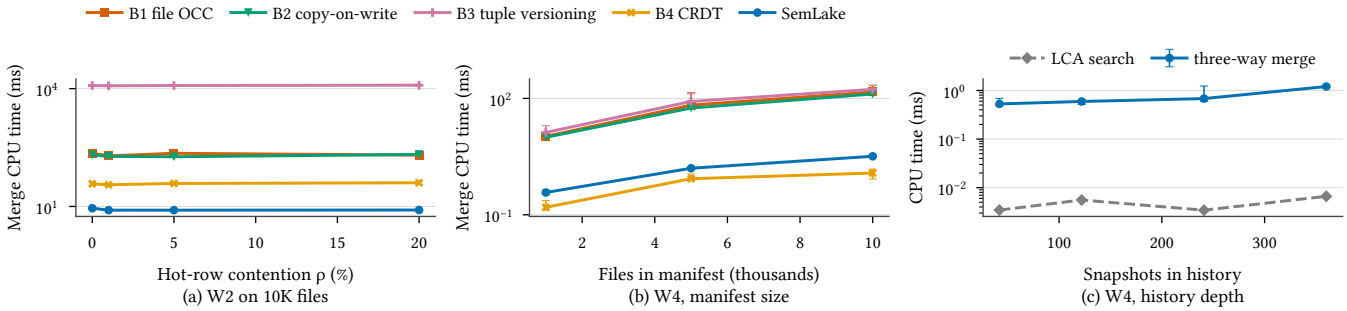


Figure 4: Three-way merge CPU latency scalability (M_2) across three lakehouse dimensions ordered by workload ($W_2 \rightarrow W_4$): (a) hotspot row contention ρ from 0% to 20% on a 10K-file table (W_2), (b) manifest depth from 1K to 10K files (W_4), and (c) commit DAG history depth from 42 to 360 snapshots (W_4). Median of 5 runs; bars show the range.

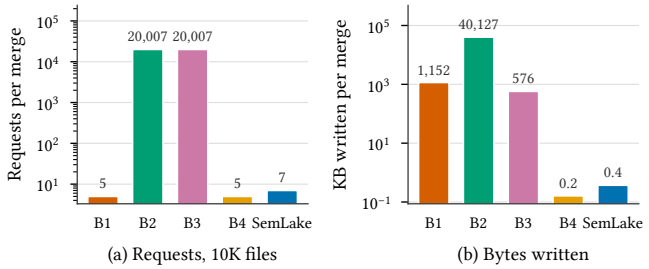


Figure 5: Object-store request count and payload bytes written (M_3) on workload W_1 at 10,000 files (B_1–B_4 vs. SemLake). SemLake issues 7 requests vs. 20,007 for Copy-on-Write (B_2) and Tuple Versioning (B_3).

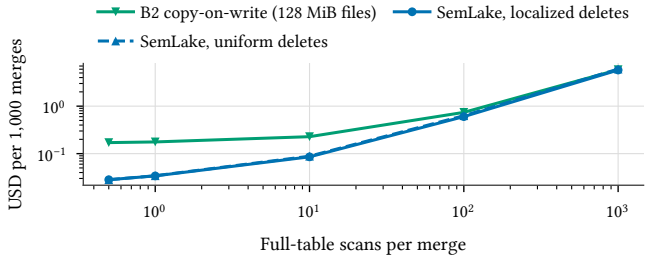


Figure 6: Total dollar cost per 1,000 merged changes (M_3, M_4: S3 API requests + 1-day storage + merge and scan CPU) against downstream full-table scans per merge, for 128 MiB base files.

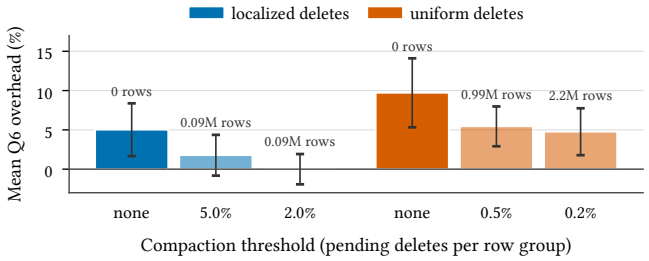


Figure 7: Mean TPC-H Q6 overhead (M_4, SF 1) over 40 merges that each delete 0.1% of the rows, by compaction threshold (fraction of a row group's rows with pending deletes). Bars show 95% intervals over the merges; labels give rows rewritten per merge.

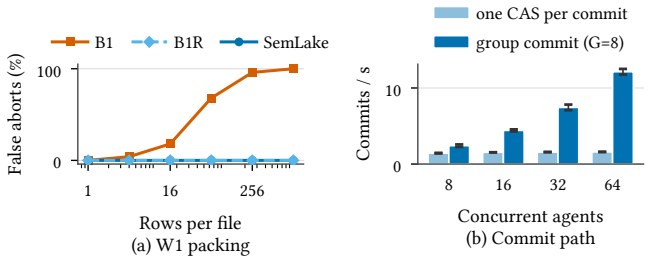


Figure 8: Ablation studies ($W_1 \rightarrow W_{\text{swarm}}$): (a) false-conflict abort rate (M_1) of B_1 vs. SemLake on W_1 as rows per Parquet file increase from 1 to 1,024; (b) applied commits per second on W_{swarm} (M_2) with one catalog CAS per commit vs. group commit ($G = 8$), 8 to 64 agents.

7.2 Semantic Merge Correctness (M_1)

Table 3 evaluates conflict recall and precision (M_1) across all baselines (B_1–B_6 and B_1R) on two benchmarks: (a) the 34 labelled merge scenarios spanning W_1 (shared-file disjoint updates), W_3 (equality, range, `BETWEEN`, `IN`, multi-column `AND`, boundary $\leq$ 3 vs. $\geq$ 3, $\neq$, unparsed `NOT/OR`, domain invariants, foreign keys, `rows_live`, and updates and inserts across or within a read range), and W_4 (additive schema evolution); and (b) 42 TPC-H [37] branch pairs (W_3) on real `lineitem` at scale factor 0.01 (60,175 rows sorted by `l_shipdate` in 31 files), pitting six refresh branches (four ship-date purges and two RF2-style deletes of order keys) against seven report branches that read with the `lineitem` filters of Q1, Q3, Q6, Q7, Q10, Q12 and Q14. A pair is a true conflict if a deleted row satisfies the report's filter; 22 pairs are. These labels, like the suite's, follow Definition 3.5, so Table 3 measures which systems enforce purge-safety, not whether a report's result changed; baselines built for other contracts score low by design.

Examining the systems in canonical order: B_1 (File OCC) tracks only physical file writes and scores 0.00 recall across both suites while suffering false aborts on W_1; the real lakeFS server decides exactly as the B_1 rule model on all 111 suite and TPC-H pairs. B_1R (Delta-style file-read tracking) catches the read-delete hazards whose rows share a file with the deleted rows (recall 0.78 on the 34-case suite, missing domain/FK invariants; recall 1.00 on TPC-H), but blocks 8 safe merges on the 34-case suite (precision 0.64) and 3 safe TPC-H pairs (precision 0.88) whose purges delete rows in files the report scanned. Row/tuple/CRDT baselines B_2, B_3, and B_4 lack

Table 3: Conflict detection on (a) the labelled scenarios and (b) the TPC-H refresh-versus-report pairs. TP and FP count reported semantic conflicts; Ab. counts write-conflict aborts, which block a merge without naming a semantic reason.

	(a) 34 scenarios, 18 conflicts					(b) 42 TPC-H pairs, 22 conflicts			
Rule	TP	FP	Ab.	Rec.	Prec.	TP	FP	Rec.	Prec.
B_1 file OCC	0	0	1	0.00	–	0	0	0.00	–
lakeFS (real)	0	0	1	0.00	–	0	0	0.00	–
B_1R Delta-style	14	8	0	0.78	0.64	22	3	1.00	0.88
B_2 copy-on-write	0	0	0	0.00	–	0	0	0.00	–
B_3 tuple versioning	0	0	0	0.00	–	0	0	0.00	–
B_4 CRDT	0	0	0	0.00	–	0	0	0.00	–
B_5 Dolt	2	0	0	0.11	1.00	0	0	0.00	–
B_6 SSI	0	0	0	0.00	–	0	0	0.00	–
SemLake, static test only	18	2	0	1.00	0.90	22	8	1.00	0.73
SemLake	18	0	0	1.00	1.00	22	0	1.00	1.00

predicate verification and score 0.00 recall on both benchmarks. B_5 (Dolt) achieves recall 0.11 (2/18) on the 34-case suite and 0.00 on TPC-H: it catches the declared `FOREIGN KEY` and `CHECK` cases, but a report's filter cannot be declared as a table constraint (W_3), and Dolt refuses to add a `CHECK` constraint that legacy rows in T_0 violate. Confirming Proposition 3.6, B_6 (Classical SSI) also scores 0.00 recall across both benchmarks: because a report branch T_B and a purge branch T_A induce only a single one-way dependency $T_B \xrightarrow{rw} T_A$ (serializable as $T_B \prec T_A$) without a back-edge $T_A \xrightarrow{rw} T_B$, SSI's dangerous-structure detector never triggers. By contrast, SemLake achieves **1.00 recall** (18/18) and **1.00 precision** on the 34-case suite, and **1.00 recall** (22/22) with **1.00 precision** on TPC-H; with the deleted-row check disabled, precision falls to 0.90 and 0.73.

Pairing the same seven report branches against 5 refresh branches that only update or insert rows (a discount fix, a reshipment, returns and new quantities on 40 orders, and an RF1-style insert of 15 orders) yields 35 more pairs, of which 10 change the report's result. SemLake catches all 10 (recall 1.00, precision 1.00); without Line 14 (`Unstable`) it catches 0. B_1R reaches recall 0.50 at precision 0.24. B_1, B_2–B_4, and B_6 catch 0; B_5 timed out after 12 minutes. At SF 0.1 (600,572 rows), all 77 pairs give SemLake recall 1.00 and precision 0.97 (B_1R: 0.79/0.60).

Result oracle. To label pairs by data rather than by Definition 3.5, we draw 30 random refreshes (deletes, updates and inserts whose clauses mix conjunctions, `NOT`, `OR` and column comparisons) and race each with random reports (`sum`, `count`, `avg` or `max` over random filters), 300 pairs in all; a pair conflicts if the report's value on the refreshed table differs from its value at the fork (105 do). SemLake misses none (recall 1.00) at precision 0.67: each of its 51 false blocks has a deleted or changed row inside the filter that leaves the value unchanged, 40 of them under `max`, which rarely moves when rows leave; without `max` reports its precision is 0.91. B_1R reaches recall 0.94 at precision 0.47, static-only SemLake precision 0.49, and the other baselines catch 0.

Figure 3 further isolates the impact of delete selectivity (W_3) on false blocks. A report reads with Q6's filter, which touches 6 of the 31 files, and a purge deletes a random range of part keys restricted to discounts Q6 does not read, so no pair is a true conflict; each point is 12 purges in three clause forms (conjunctive, `NOT` and `OR`), deleting from 0.019% to 20% of the table. B_1R blocks the report as soon as the purge touches one of the files Q6 reads: its false-block rate is 1.00 at 0.1% of rows. Static-only SemLake blocks the `NOT` and `OR` forms (rate 0.67), whereas full SemLake has a **0.00 false-block rate** at every selectivity.

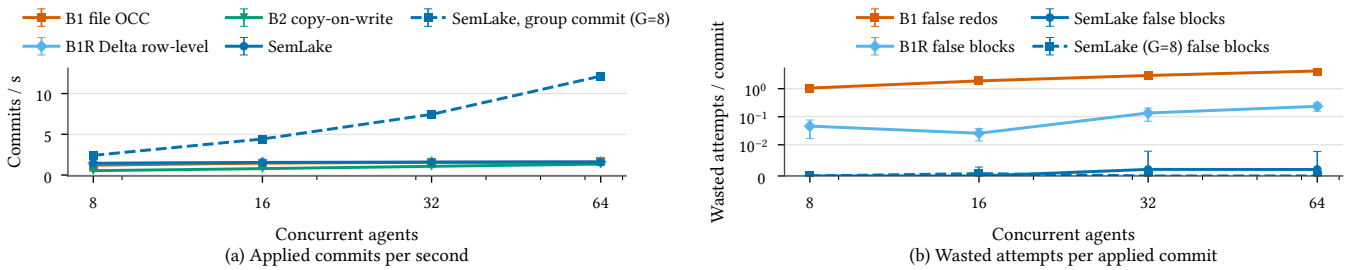
Key Insight (M_1): *Predicate-level verification over commit summaries eliminates silent read-delete, read-update and domain corruption (W_3) with 1.00 recall, avoiding both the low recall of B_1, B_2–B_6 and the false blocks of file-granularity read tracking (B_1R).*

7.3 Merge Latency and Metadata Scalability (M_2)

Figure 4 evaluates three-way merge CPU latency (M_2) across three scaling dimensions ordered by workload ($W_2 \rightarrow W_4$): (1) **Hotspot row contention (W_2, Figure 4a):** on a 10,000-file table, SemLake merges in 9.02 ms at $\rho = 0$ and 8.12 ms at $\rho = 20\%$, where colliding cells become VSA choice nodes; B_1 takes 227.3 ms and 203.2 ms because it compares full file lists. B_4 takes 37.9 ms at $\rho = 0$ by skipping conflict checks and overwriting collisions via last-writer-wins. (2) **Manifest depth (W_4, Figure 4b):** as the manifest grows from 1K to 10K Parquet files, SemLake merges a schema change and an append in 0.38 ms and 3.22 ms, while B_1, B_2, and B_3 take 147.0 ms, 129.8 ms, and 171.4 ms at 10K files, and B_3 issues 20,009 object-store calls to rebuild its dictionaries. (3) **Commit DAG history depth (W_4, Figure 4c):** from 42 to 360 snapshots, generation-pruned LCA discovery takes 3.5 and 6.6 μs while end-to-end merge CPU rises from 0.53 ms to 1.20 ms: each merge replays one commit per side, but clones the growing in-memory store, lists `manifests/`, and carries forward one uncompacted `delta-<snap>` descriptor per prior merge. Under the cloud latency model of Section 6, a W_2 merge at $\rho = 0$ is estimated at 0.55 s for SemLake and 0.48 s for B_1, because SemLake reads two commit summaries and writes three metadata objects plus one delta file. Summaries grow with a purge because they carry pre-images: deleting 10% of SF 0.1 `lineitem` (59,790 rows) writes a 15.6 MB summary (12.1 MB of pre-images, 9.2 MB if only filter columns were kept), and a merge that checks it against a report takes 1.7 s, against 0.40 s at 0.1%; no merge in our experiments reads a base row.

7.4 Cloud Object-Store Cost Efficiency (M_3)

Figure 5 evaluates S3 request counts and write amplification (M_3) on W_1 across a 10,000-file table where two branches update disjoint rows across 5,000 files each. SemLake completes the merge in **7 S3 requests** (3 `PUT`, 3 `GET`, 1 `LIST`), writing 380 bytes of metadata ($0.0000212) and rewriting no base file. Copy-on-Write B_2 reads and rewrites all 10,000 modified files, issuing 20,007 S3 requests (10,001 `PUT`, 10,005 `GET`, 1 `LIST`) at $0.054012, a **99.96% S3 request-cost reduction** for SemLake. B_1 issues only 5 requests because it aborts the merge. Similarly, on W_4, SemLake issues 7 S3 requests ($O(\Delta_{\text{manifest}})$) compared to 20,009 requests in B_3. On a live Amazon S3 bucket (`eu-north-1`; 3 runs per point) request counts match the in-memory counts exactly. SemLake merges W_1 in 1.68 s at 1K files and 2.15 s at 10K, and W_4 in 1.52 s and 1.52 s; B_2 issuing 32 requests at a time takes 16.2 s and 148.6 s on W_1 (69× SemLake at 10K), and 469 s at 1K one request at a time. Our client's requests take longer than the model's 30–100 ms, so the model underestimates these times 3.4–4.5×; recalibrated with measured medians (PUT 297, GET 154, LIST 147 ms) it predicts every sequential run to within 1.01–1.40×, the high end being the 10K-file W_1 merge, whose



(a) Applied commits per second

(b) Wasted attempts per applied commit

Figure 9: Extension case study on agent swarms (W_{swarm}): (a) applied commits per second (M_2) from 8 to 64 concurrent agents (B_1, B_1R, B_2 vs. SemLake; with measured S3 latencies, one CAS per commit levels off near 1.6 commits/s, while SemLake with group commit ($G = 8$) reaches 12.1 commits/s at 64 agents; 5 seeds, 95% intervals); (b) wasted agent work (M_1: false transaction redos in B_1 and false read blocks in B_1R) per applied commit.

Table 4: TPC-H Q6 on SF 1 `lineitem` after a merge left deletes in a DV: median time of 5 processes, and overhead with a 95% confidence interval.

Deletes	Rows	Clean groups	Base (ms)	DV (ms)	Overhead (95% CI)
localized	1%	60/61	190.5	198.2	+0.9% (−3.4 to +6.9%)
uniform	0.1%	0/61	194.0	206.8	+4.2% (−1.7 to +12.5%)
uniform	1%	0/61	187.4	207.2	+6.0% (−5.3 to +23.1%)
uniform	5%	0/61	178.6	187.8	+5.4% (+0.2 to +14.2%)

two 0.74 MB summaries the model does not charge for transfer. A merge on S3 thus takes over a second, short of M_2's sub-second goal: the time is round trips, not CPU.

While keeping deletion vectors uncompacted minimizes merge-time write I/O (M_3), it introduces bitmap filtering work during subsequent analytical reads (M_4). Validating the crossover bound N^*_{scans} from Section 5.2, Figure 6 models total dollar cost per 1,000 merges (combining the measured S3 API requests, one day of S3 storage for written bytes, 128 MiB Parquet file rewrites for B_2, and CPU at $0.10/hour, using the scan overheads of Table 4) against downstream full-table scans per merge: (1) at 10 scans per merge, 1,000 merges with open DVs cost $0.086 in SemLake for localized deletes ($0.088 for uniform deletes) vs. $0.228 for CoW B_2; (2) leaving DVs open stays cheaper than rewriting until **3,094 scans per merge** when deletes are localized to a single row group (W_1), and **446 scans per merge** when uniform deletes dirty every row group. Past that point the merged files should be compacted (Section 7.5).

7.5 Post-Merge Analytical Scan Overhead (M_4)

Table 4 reports the post-merge read penalty (M_4) when running TPC-H Q6 [37] in DuckDB (one thread) over real SF 1 `lineitem` (6,001,215 rows in 61 row groups of 100,000 rows) after a merge left a DV open. Clean row groups with empty deletion vectors ($G_{\text{rg}} - G_{\text{dirty}}$) are scanned directly from untouched base files; dirty row groups (G_{dirty}) apply a positional boolean mask join.

Under localized updates (W_1, where 1% of the rows are deleted inside one row group, $G_{\text{dirty}}/G_{\text{rg}} = 1/61$), the median scan overhead is **+0.9%** (95% CI: −3.4% to +6.9%, within our < 8% target). When the same deletes are spread over every row group ($G_{\text{dirty}}/G_{\text{rg}} = 1$), every group needs the positional mask join and median overhead rises to +6.0%. At SF 10 (59,986,052 rows across 600 row groups, base scan 1,714 ms), overheads are −1.4% (−5.6% to +2.7%) localized and +5.6% (+2.9% to +9.0%) uniform, measuring SQL-level mask joins without a native bitmap reader.

Figure 7 traces a table through 40 merges, each deleting 0.1% of its rows, and measures Q6 after every merge under row-group compaction thresholds τ. Without compaction, mean overhead is 5.0% for localized deletes and 9.7% for uniform ones. For localized deletes, $\tau = 2\%$ brings mean overhead to 0.0% while rewriting 89,549 rows per merge (under one row group). For uniform deletes, the best threshold ($\tau = 0.2\%$, 4.8% mean overhead) rewrites 2,233,892 rows per merge (over a third of the table) because all row groups cross τ nearly together, showing that threshold compaction suits localized edits while uniform deletes favor batch compaction.

7.6 Ablation Studies

We isolate five core design choices in SemLake ordered by workload ($W_1 \to W_3 \to W_4 \to W_{\text{swarm}}$): (1) **Parquet file packing density ($W_1 \to M_1$, Figure 8a):** two branches each update 4 random rows of a 1,024-row table; as rows packed per Parquet file increase from 1 to 1,024, B_1's false-abort rate climbs from 0 to 1.00, whereas B_1R and SemLake maintain a **0% false-abort rate** regardless of file size because both compare rows. (2) **Tier-2 deleted-row check ($W_3 \to M_1$, Figure 3):** disabling `AnyMatch` in Algorithm 2 leaves recall unchanged but lowers precision from 1.00 to 0.90 on the 34-case suite and from 1.00 to 0.73 on the TPC-H pairs, and raises the false-block rate for the `NOT` and `OR` forms in Figure 3 from 0.00 to 0.67. (2b) **Update check ($W_3 \to M_1$):** without Line 14, recall falls to 0.78 on the suite and 0.00 on the update pairs; delete pairs are unchanged. (3) **Exact DAG replay since the fork ($W_4 \to M_1$):** tip-only replay loses the predicates of earlier merges and lets through a report merged once before it meets a purge, which Algorithm 1 blocks. (4) **Catalog commit-path batching ($W_{\text{swarm}} \to M_2$, Figure 8b):** with a 10 ms catalog round trip, one CAS per commit reaches 1.6 commits/s at 64 agents, whereas group commit ($G = 8$) reaches **12.1 commits/s**.

7.7 Extension Case Study: Concurrent Agent Swarms (W_{swarm})

Finally, we evaluate SemLake on the agent-swarm extension W_{swarm} [4, 20, 28, 41] across M_1–M_3 in a discrete-event simulation. N agents repeatedly branch from `main`, work for a random time (mean 2 s), and commit one task: an update of four rows (80%), a range delete (5%), or a report that reads a range and writes an aggregate (15%), on a table of 32 files of 32 rows. Publishing is a CAS with a 10 ms catalog round trip. SemLake runs on the real engine, and each merge advances simulated time by its measured CPU time plus its

Table 5: Simulated swarm, 60 s per run, mean of 5 seeds (± 95% CI). Haz.: hazards committed and caught, summed over seeds. Blocks and redos: false semantic blocks and false file-conflict redos per run.

N	Rule	Commits/s	Haz.	Caught	Blocks	Redos
8	B_1 file OCC	1.21 ± 0.07	0	0	0.0	76.0
8	B_1R Delta-style	1.44 ± 0.04	0	0	4.0	0.0
8	B_2 copy-on-write	0.54 ± 0.05	0	0	0.0	0.0
8	SemLake	1.43 ± 0.04	0	0	0.0	0.0
8	SemLake, group commit	2.42 ± 0.16	0	4	0.0	0.0
32	B_1 file OCC	1.56 ± 0.02	0	0	0.0	278.0
32	B_1R Delta-style	1.59 ± 0.01	1	3	13.0	0.0
32	B_2 copy-on-write	1.07 ± 0.12	1	0	0.0	0.0
32	SemLake	1.58 ± 0.01	0	4	0.2	0.0
32	SemLake, group commit	7.45 ± 0.37	0	15	0.0	0.0
64	B_1 file OCC	1.61 ± 0.01	3	0	0.0	413.2
64	B_1R Delta-style	1.62 ± 0.00	5	10	22.8	0.0
64	B_2 copy-on-write	1.34 ± 0.09	0	0	0.0	0.0
64	SemLake	1.60 ± 0.01	0	21	0.2	0.0
64	SemLake, group commit	12.14 ± 0.37	0	43	0.0	0.0

counted requests priced at our measured S3 medians; the baselines are rule models charged with the same model. A report and a delete that ran concurrently over shared rows are a hazard if both commit. Runs last 60 s; a purge-heavy mix (15% deletes, 30% reports) runs at 8 and 32 agents, and hazard totals cover both mixes.

B_1 redoes work on every file overlap and B_2 rewrites every touched file. B_1R and SemLake both avoid file conflicts, but B_1R blocks a report whenever a concurrent delete touched a file it read: 13.0 blocks per run at 32 agents, against 0.2 for SemLake. Over all runs, B_1 committed 15 hazards, B_2 8, B_1R 11, and SemLake **0**. B_1R's read check is one-directional: a delete that commits after a report read nothing the report wrote. Group commit ($G = 8$) raises SemLake from 1.6 to **12.1 commits/s** at 64 agents (Figure 9), as each lost race costs another merge (450 vs. 63 requests per commit).

8 RELATED WORK

Lakehouse Storage and Row-Level Operations. Delta Lake [6], the Lakehouse paradigm [40], and comparative storage studies [10, 21, 23] established metadata-driven ACID transactions over cloud object stores. Okolnychyi et al. [31], Apache Iceberg v3 [5], and Delta Lake deletion vectors [27] introduced Roaring DVs and row lineage for single-branch MoR updates, while Apache Hudi [35] and Lance [11] provide multi-modal metadata and point-lookup indices. Durner et al. [15] quantified the S3 request-cost model that informs our $O(\Delta_{\text{manifest}})$ delta-manifest design. LakeVilla [19], Götz et al. [18], and TreeCat [30] coordinate multi-table commits and catalog trees, but omit cross-branch predicate conflict verification.

Dataset Versioning and Branching. Dataset versioning principles [8], DataHub [7], Decibel [29], OrpheusDB [22], and ForkBase [39] pioneered storage layouts for branched datasets. Bauplan [36] provides zero-copy branch references for cloud WAP pipelines, and Dolt [14] performs cell-level merges over Prolly trees with SQL schema constraints, missing query read predicates and partial-domain invariants.

Program Merge and Version-Space Algebra. Sousa et al. [34] formalized verified three-way program merge over ASTs, Zhu and He [43] introduced VSA to represent conflicting AST candidates, and Zhu et al. [42] combined top-down and bottom-up traversals. Subsequent work detects behavioral conflicts via unit tests [12] or static analysis [13]. SemLake brings three-way merge and VSA from ASTs to cloud lakehouses by lifting verification to SQL commit predicates and Roaring DVs.

Concurrency Theory and Replicated Views. Eswaran et al. [16] introduced predicate locks and Jordan et al. [24] precision locks, which test written tuples against readers' predicates; optimistic schemes [25, 26] validate read and scan sets at commit. SemLake applies the same test between long-lived branches, from stored delete-time values, after both have committed. Adya [1] formalized multi-version dependency graphs, and Cahill et al. [9] developed Serializable Snapshot Isolation (SSI). Proposition 3.6 shows that purge-safe branch merging and two-branch serializability are incomparable, because both branches' outputs coexist in T_M. CRDTs [2] and CRDV [17] guarantee automatic state convergence without branch isolation or semantic conflict blocking.

Extensions to Agentic Systems. Git4Data [20] and Chronos [41] provide copy-on-write and bolt-on branching for AI agents, BranchBench [4] evaluates branching vs. query latency under agentic workloads, and ATR [28] studies selective revalidation for long-running agents. SemLake adds the missing three-way semantic merge to these substrates.

9 LIMITATIONS AND CONCLUSION

Conducting our study on a single academic workstation required combining physical storage execution with analytical and discrete-event simulation, which introduces five boundaries for future work: (1) *Cloud and baseline modeling*: Parquet I/O, Roaring bitmaps, DuckDB scans (SF 1 and SF 10), Dolt and lakeFS merges are physical, and our S3 runs (`eu-north-1`, 1K and 10K files) confirm the request counts and calibrate the latency model, and the swarm simulator uses the measured request latencies, but the other 10K-file estimates still use the uncalibrated model; B_1R, B_2, and B_6 remain rule models (only B_1 is checked against a real system), and Nessie [32] is future work. (2) *Predicate scope*: the checks compare rows rather than query outputs, blocking changes that leave an aggregate unchanged (precision 0.67 under the result oracle), while joins and subqueries fall outside single-table row checks. (3) *Prototype scale and scan path*: our Python prototype keeps branch row deltas in memory (TPC-H merge pairs up to SF 0.1), applies DVs via a DuckDB positional mask join rather than a native bitmap reader, and retains one delta descriptor per uncompacted merge. (4) *Multi-table merges*: extending group commit to multi-table lakehouse transactions [18, 19] remains open. (5) *Declared reads and pre-images*: soundness requires declared reads, and summaries keep about 202 bytes of pre-image per deleted row, which also retains erased values until the summaries are pruned.

SemLake brings three-way semantic merge, backed by a soundness proof, to branched data lakehouses. By decoupling two-tier predicate and row-value verification over compact commit summaries from zero-copy physical reconciliation over Roaring deletion vectors, SemLake blocks the semantic conflicts that file- and row-level merges miss, with object-store requests that scale with the branch delta.

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